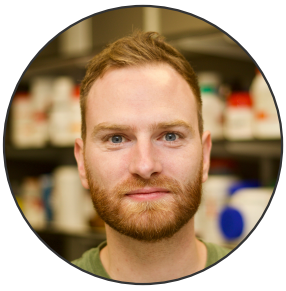

Conor M. **Finlay**

Senior Research Fellow

Trinity Translational Medicine Institute, Trinity College Dublin

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Cellular Immunologist and bioinformatician combining molecular, in vivo and translational research.

Senior Research Fellow, Trinity College Dublin (2022-present)

- SFI-Pathways Programme Fellow (PI, starting August 2024)
- TCD Omics - Single cell RNA-sequencing core service (founder and scientific director), since 2021
- HEALED consortium - molecular biology and sequencing lead (with Aideen Long), since August 2023
- COVID-19 SFI-funded strategic partnership - transnational immunology ,up to mid-2023
- Trinity Kidney Centre - lab management, student supervision(with Mark Little), up to mid-2023
- Multi-modal clinical data analysis (with Mark Little), up to mid-2023
- Co-supervising longitudinal immune cell phenotyping study, PARADISE project, (with Mark Little), 2022-present

Honorary Research Fellow, University of Manchester (2021-present)

- Collaboration on *MRC* programme grant with Prof. Judith Allen.

Past roles & Education

2021-2022	Research Fellow, Supervisor: Prof Mark Little	*Trinity*
2020	Informatics Training Scheme (Wellcome Trust TPA),Programming and Computational Approaches to Biology Modules	*University of Manchester*
2017-2021	Research Associate, Supervisor: Prof Judith Allen.	*University of Manchester*
2016-2017	Lecturer (teaching relief post), School of Biochemistry and Immunology	*Trinity*
2013-2016	Post-doctoral Researcher, Supervisor: Prof Kingston HG Mills.	*Trinity*
2013	PhD in Immunology, Immune modulation by the helminth Parasite Fasciola hepatica	*Trinity*
2007	BA(Mod) in Natural Sciences, Major: Biochemisty with Immunology, Grade: 1:1	*Trinity*

Funding

I have received a total funding of 684,106 Euro in funding to my lab in Trinity. In the UK I co-wrote a programme grant (as Research Co-Investigator) on which I now collaborate.

SFI-Pathways Programme — *Science Foundation Ireland*
594K, PI — *2024-2028*
- Macrophage Activation and Transition in the Autoimmune Kidney (MacATAK): uncovering functional outcomes of macrophage differentiation in the inflamed kidney

Trinity Doctorate Award 2024-2025, PI-led application — *Trinity Doctorate Award*
122K, PI — *2024-2028*
- Mapping Macrophage Cell State Transition in Inflammation and Infection at Single Cell Resolution

The Meath Foundation 2024 — *The Meath Foundation*
70K, Co-I (35K to my group) — *2024-2026*
- 'Characterisation and Inhibition of Monocyte-Driven Inflammation in AAV (CIMDIA)'

Higher Education Research Equipment Grant — *Higher Education Authority*
30.2K, PI — *2024-2025*
- Cell counting solution to support TCD Omics

Dean's Research Initiatives Fund — *Faculty of Health Sciences, TCD*
20K, PI — *2024*
- Mapping human macrophage anti-microbial function in pleural infection

Enterprise Ireland-funded Industry collaboration — *Legend Biotech*
93K — *2024*
- Title withheld, scRNA-sequencing of blood cancer

Building Engagements in Health Research Scheme — *Internal TCD*
10K Eur, Co-PI — *2023*
- Understanding mast cells and their progenitors in Telangiectasia Macularis Eruptiva Perstans (TMEP)'

MR/V011235/1 Programme Grant *Medical Research Council*

2.26 Million GBP, Research Co-Investigator *2021*

- "Macrophages in type 2 immunity: unravelling susceptibility and resistance to tissue nematode infection". I co-wrote grant and provided scientific design and preliminary data. Awarded an honorary position at UoM to facilitate ongoing collaboration after leaving Manchester

Building Engagements in Health Research Scheme *Internal TCD*

10K Eur, Co-PI *2021*

- 'Nanoparticle modulation of neutrophil and monocyte responses to ANCA'

Presentations, Awards & contributions

Presentations

- 4 invited speaker presentations
- 13 conference oral presentations, selected from abstract, * 8 conference poster presentations

Awards and media

- Commentry in *Immunity*: "Macrophages show up in style when Th2 lymphocytes organize their homecoming"
- Paper featured on The Immunology Podcast - Episode 51
- Immunity paper featured on August 2023 edition of *Journal of Allergy and Clinical Immunology* News Beyond Our Pages.

Discipline Contributions

- Reviewed articles for 6 journals.
- Reviewed grants for 4 funding bodies
- Sat on 3 researcher interview panels
- Co-Inventor on a patent for Trinity Spin-out (Parvalis Tx)

Research leadership and collaborations

Student projects 16 projects designed and supervised across Trinity (12-15 weeks) and Manchester (6 months). Currently co-suppervise a IRC-funded PhD student.

TCD Omics Core facility. I have established formal pipelines for a single cell RNA-sequencing core service 'TCD Omics'.

16 active collaborations, with groups in NIH, Leiden, Manchester, RCSI, Trinity college Dublin, St James' hospital.

Selected publications (last 3 years)

1. Ridge, K., Moran, B., Alvarado-Vazquez, P. A., Hallgren, J., Little, M. A., Irvine, A. D., O'Farrelly, C., Dunne, J., Finlay (co-senior), C. M., & Conlon (co-senior, N. (2024). Lin − CD117 + CD34 + FceRI + progenitor cells are increased in chronic spontaneous urticaria and predict clinical responsiveness to anti- IgE therapy. *Allergy*. `https://doi.org/10.1111/all.16127`
2. Han, J., Gallerand, A., Erlich, E. C., Helmink, B. A., Mair, I., Li, X., Eckhouse, S. R., Dimou, F. M., Shakhsheer, B. A., Phelps, H. M., Chan, M. M., Mintz, R. L., Lee, D. D., Schilling, J. D., Finlay, C. M., Allen, J. E., Jakubzick, C. V., Else, K. J., Onufer, E. J., … Randolph, G. J. (2024). Human serous cavity macrophages and dendritic cells possess counterparts in the mouse with a distinct distribution between species. *Nature Immunology*, *25*(1), 155–165. `https://doi.org/10.1038/s41590-023-01688-7`
3. Tachó-Piñot, R., Stamper, C. T., King, J. I., Matei-Rascu, V., Richardson, E., Li, Z., Roberts, L. B., Bassett, J. W., Melo-Gonzalez, F., Fiancette, R., Lin, I.-H., Dent, A., Harada, Y., Finlay, C., Mjösberg, J., Withers, D. R., & Hepworth, M. R. (2023). Bcl6 is a subset-defining transcription factor of lymphoid tissue inducer-like ILC3. *Cell Reports*, *42*(11), 113425. `https://doi.org/10.1016/j.celrep.2023.113425`
4. Finlay, C. M., Parkinson, J. E., Zhang, L., Chan, B. H. K., Ajendra, J., Chenery, A., Morrison, A., Kaymak, I., Houlder, E. L., Murtuza Baker, S., Dickie, B. R., Boon, L., Konkel, J. E., Hepworth, M. R., MacDonald, A. S., Randolph, G. J., Rückerl, D., & Allen, J. E. (2023). T helper 2 cells control monocyte to tissue-resident macrophage differentiation during nematode infection of the pleural cavity. *Immunity*, *56*(5), 1064–1081.e10. `https://doi.org/10.1016/j.immuni.2023.02.016`
5. McEntee, C. P., Houston, S., Finlay, C. M., Rossi, S., Liu, G., Shaw, T. N., Casulli, J., Fife, M., Smedley, C., Griffith, T. S., Pepper, M., Hussell, T., Hansbro, P. M., Schwartz, J.-M., Paidassi, H., & Travis, M. A. (2023). A subset of CD4+ effector memory T cells limit immunity to pulmonary viral infection and prevent tissue pathology via activation of latent TGFβ. *bioRxiv*, 2023.03.02.527395. `https://doi.org/10.1101/2023.03.02.527395`
6. Fiancette, R., Finlay, C. M., Willis, C., Bevington, S. L., Soley, J., Ng, S. T. H., Baker, S. M., Andrews, S., Hepworth, M. R., & Withers, D. R. (2021). Reciprocal transcription factor networks govern tissue-resident ILC3 subset function and identity. *Nature Immunology*, *22*(10), 1245–1255. `https://doi.org/10.1038/s41590-021-01024-x`
7. Cunningham, K. T., Finlay, C. M., & Mills, K. H. G. (2021). Helminth Imprinting of Hematopoietic Stem Cells Sustains Anti-Inflammatory Trained Innate Immunity That Attenuates Autoimmune Disease. *Journal of Immunology (Baltimore, Md. : 1950)*, *206*(7), 1618–1630. `https://doi.org/10.4049/jimmunol.2001225`

8. Finlay, C. M., Cunningham, K. T., Doyle, B., & Mills, K. H. G. (2020). IL-33-Stimulated Murine Mast Cells Polarize Alternatively Activated Macrophages, Which Suppress T Cells That Mediate Experimental Autoimmune Encephalomyelitis. *Journal of Immunology (Baltimore, Md. : 1950)*, *205*(7), 1909–1919. https://doi.org/10.4049/jimmunol.1901321
9. Finlay, C. M., & Allen, J. E. (2020). The immune response of inbred laboratory mice to Litomosoides sigmodontis: A route to discovery in myeloid cell biology. *Parasite Immunology*, *42*(7), e12708. https://doi.org/10.1111/pim.12708

Referee

Prof Judi Allen, Professor of Immunobiology; Lydia Becker Institute University of Manchester

Up to date long-form CV availible on GitHub (here)